Computers and Education: Artificial Intelligence 10 (2026) 100590

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Teacher feedback and ChatGPT feedback on Chinese university EFL learners' English essay revision: A mixed-methods study

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ARTICLE INFO

Keywords:
Chinese university EFL learners
Teacher feedback
ChatGPT feedback
Revision in FL writing
Perceived usefulness
Perceived ease of use
Benefits and drawbacks

ABSTRACT

Adopting mixed methods, the study compared teacher feedback and ChatGPT feedback on 35 Chinese university EFL learners' essay revision, their Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) and their perceived benefits and drawbacks of the feedback. Students wrote two English essays and received writing feedback from their English teacher and ChatGPT respectively. They revised one essay using teacher feedback and another using ChatGPT feedback. Upon revision, students responded to a 5-point Likert-scale questionnaire regarding PU and PEOU of the two types of feedback. The results showed that students obtained significantly higher scores on the revised essays using ChatGPT feedback. However, students perceived more useful and easier to use than ChatGPT feedback in the revision process. Thematic analysis of interviews revealed that while students valued teacher feedback for its emotional support and clarity, they found ChatGPT feedback more comprehensive yet often linguistically complex and generic. Students expressed mixed feelings, acknowledging distinct strengths and limitations of each feedback type.

1. Introduction

Writing proficiency is a crucial yet challenging skill for students learning English as a Foreign Language (EFL). Providing timely, detailed, and actionable feedback on learners' writing is one of the most effective ways to facilitate improvement. However, delivering such feedback is both time-consuming and labor-intensive, placing a significant burden on EFL instructors, especially in large classes commonly found in contexts like China (Meyer et al., 2023). This practical constraint often limits the depth and personalization of feedback that teachers can provide.

The emergence of generative artificial intelligence (GAI), particularly large language models (LLMs) like ChatGPT, may present a potential paradigm shift. ChatGPT can generate coherent, context-aware text and provide detailed analyses on various aspects of writing, offering a promising, scalable alternative or supplement to traditional teacher feedback. Its potential to provide instant, comprehensive, and individualized writing assistance has sparked considerable interest among educators and researchers.

However, the integration of GenAI tools like ChatGPT into pedagogical practices must be guided by empirical evidence. While initial studies have explored ChatGPT's capabilities in assessment and feedback generation, comparative research examining its efficacy against human teacher feedback in authentic EFL writing revision contexts shows inconsistent findings. Some studies suggest ChatGPT feedback can lead to significant improvements in revision quality (Myer et al., 2023), while others find no significant difference compared to teacher feedback on longer-term proficiency development (Escalante et al., 2023). Furthermore, students' perceptions, encompassing their sense of the feedback's usefulness, ease of use, and affective perceptions, are critical factors influencing the successful adoption of any new educational technology (Han & Guo, 2025), yet these dimensions are not fully understood in the context of GenAI versus teacher feedback.

Therefore, a nuanced, mixed-methods investigation is needed to move beyond a simple comparison of revision outcomes and understand the complex interplay between feedback effectiveness and learner perception. The present study aims to fill this gap by systematically comparing: 1) the impact of teacher feedback vs. ChatGPT feedback on Chinese university EFL learners' quality of essay revision; 2) their Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) regarding these two types of feedback; and 3) the qualitative perceptions of the benefits and drawbacks of each feedback type.

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https://doi.org/10.1016/j.caeai.2026.100590
Received 15 July 2025; Received in revised form 28 March 2026; Accepted 29 March 2026
Available online 3 April 2026

2. Literature review

2.1. ChatGPT, GAI, and LLMs

ChatGPT, a type of GAI developed by OpenAI, was launched on November 30, 2022 (OpenAI, 2023). Its origins can be traced back to the emergence of LLMs in 2018, marked by the release of BERT (Jawahar et al., 2019). Since then, various LLMs, including GPT (OpenAI, 2023), have been introduced. The term "large" highlights the vast scale of text data these models are trained on, often comprising millions to billions of words (Brown et al., 2020). This extensive training enables the model to understand the complexities of language, producing fluent and coherent text resembling human communication. Although the technical specifics of LLMs differ, they generally rely on deep learning techniques. This involves training artificial neural networks on massive datasets to predict and generate words in a sequence based on preceding text. As a result, these models can create text that flows naturally and mimics human language.

ChatGPT gained significant attention not just because it introduced an innovative concept but also because it was the first of its kind to be broadly accessible to the general public without requiring technical expertise, such as coding skills. Its chat-style interface allows users to interact with the model in plain language. By simply logging onto a website, users can input a prompt and receive near-instant responses. Additionally, ChatGPT can maintain conversational context, enabling it to provide relevant replies that build on previous interactions.

The capabilities of ChatGPT and similar LLMs are remarkable, covering tasks such as writing different text types; offering critiques and edits; summarizing or expanding texts; adapting content of writing to different styles or genres; generating code from natural language descriptions; creating lesson plans and assessment strategies for teachers; and providing personalized tutoring for students (Kalla et al., 2023).

2.2. ChatGPT in writing

An important applied area of ChatGPT is in different types of writing. Existing research has been conducted on using ChatGPT in professional writing, such as in the disciplines of business writing (Cardon et al., 2023); healthcare communications, like drafting radiology reports (Lyu et al., 2023), patients' answers (Wu et al., 2024), surgical case reports (Ho et al., 2023); academic writing, including writing abstracts for conferences (Babl & Babl, 2023) and research proposals (Athaluri et al., 2023); as well as assessing and grading writing.

Past investigations have compared the agreement of the ChatGPT and human raters in terms of assessing essays (Dai et al., 2023; Mizumoto & Eguchi, 2023; Parker et al., 2023). While some studies found that ChatGPT was stricter in terms of grading (Parker et al., 2023), others concluded a high consistency between ChatGPT and human raters' assessment results (Mizumoto & Eguchi, 2023).

Beyond its functions in grading and assessment, one of ChatGPT most notable capabilities is to provide writing feedback (Sallam, 2023). ChatGPT can deliver immediate and tailored critiques on students' written work, addressing elements such as structure, content, grammar, vocabulary, and mechanics. This feature positions ChatGPT as a potential personalized writing assistant for individual EFL learners (Barrot, 2023; Steiss et al., 2024).

2.3. ChatGPT feedback on cognitive outcomes of FL writing

Research has examined the effects of ChatGPT feedback on FL writers' writing competence and quality of revision) and affective (e.g., motivation, emotions, and satisfaction) outcomes of FL writing (Escalante et al., 2023; Myer et al., 2023; Song & Song, 2023; Teng, 2024). Some studies compared ChatGPT feedback versus no feedback, whereas others compared ChatGPT feedback with teacher feedback or peer feedback.

Concerning ChatGPT on cognitive outcomes of FL writing, Myer et al. (2023) compared quality of argumentative essay revision and performance on new argumentative writing task between German EFL learners who received either no feedback or feedback generated by ChatGPT. Although there were no significant differences on new task writing scores, ChatGPT group obtained significantly higher scores of the revised essays than no feedback group. This result was understandable, as without receiving any feedback, EFL learners might not even realize the problems in their essays, hence would not be able to make improvement in the revision. As this research only involved ChatGPT feedback, it is unknown about how ChatGPT feedback compared with other types of feedback on FL writing.

Song and Song (2023) compared Chinese university EFL learners' writing competence development in two conditions: writing practice with English teacher or with ChatGPT in a 12-weeks longitudinal study. The results showed that students who practiced writing with ChatGPT had significantly better writing competence that those practiced writing with English teacher. In this study, however, students not only received writing feedback from either teacher or ChatGPT, but also completed various learning activities designed to improve their English proficiency. These activities might also have improved students' grammar, vocabulary usage, and sentence structure, all of which are building blocks for skilled writing. Hence, the findings of the study might have been affected by the learning activities.

To eliminate the confounding factor brought by learning activities and exercises, Escalante et al. (2023) designed a quasi-experiment in which two classes EFL learners either received ChatGPT feedback or teacher feedback on their academic writing for six weeks. Students' academic writing proficiency was measured in pre- and post-tests. No significant differences were found in pre- and post-writing proficiency between the two groups. The researchers attributed the non-significant results to the short intervention period as writing proficiency might take longer time to develop. However, it is unknown how effective the two types of feedback on students' essay revision as this study did not measure quality of revision. The present study will compare the two types of feedback (i.e., teacher feedback and ChatGPT feedback) on EFL learners' quality of essay revision, which constitutes the first aim of the study.

2.4. ChatGPT feedback on affective outcomes of FL writing

With regard to how ChatGPT feedback may impact the affective aspects in FL writing, the findings to date were not conclusive. On the one hand, studies reported that EFL learners had more positive affective responses when interacting with ChatGPT feedback. For instance, Myer et al. (2023) found that the ChatGPT group exhibited higher task values and more positive emotions during essay revision than the no-feedback group. Similarly, Teng (2024) also showed that Chinese university EFL learners receiving ChatGPT feedback had higher ratings on motivation, self-efficacy, engagement, and collaborative writing than their peers receiving teacher feedback.

However, in a mixed-methods study comparing EFL learners' satisfaction of ChatGPT feedback and teacher feedback, Escalante et al. (2023) reported no significant differences of satisfaction between ChatGPT feedback and teacher feedback as demonstrated by the quantitative results. Students' preference for ChatGPT feedback or teacher feedback was also a near even-split. The qualitative comments revealed that students liked the personal interaction of teacher feedback, whereas they were positive about the clarity, consistency, and specificity of ChatGPT feedback on academic writing style and vocabulary. It should be noted that this study adopted a within-subjects design, this means that the same learners' satisfaction and preferences of the ChatGPT feedback and teacher feedback were compared. This differed from Teng's (2024) study, in which the two types of feedback were given to different participants. Clearly, more studies are required to examine the impact of feedback type on learners' affection in FL revising process. To provide additional empirical evidence, the second aim of the present study was to compare EFL learners' perceptions of the teacher versus ChatGPT feedback in terms of the usefulness (PU) and ease of use aspects (PEOU).

PU and PEOU are the two core factors pertaining to users' motivation in the technology adoption literature (Davis et al., 2024). PU is conceived as the extent to which an individual perceives that using a technology or an information system would improve the performance in a specific task; whereas PEOU is defined as the extent to which an individual perceives that using a technology or an information system would be effort-free (Davis, 1989). Research has collectively shown that both PU and PEOU affect students' acceptance and adoption of many emerging technologies in learning (Scherer et al., 2019), including AI (Zhang et al., 2023). Therefore, when comparing students' perceptions of teacher and ChatGPT feedback, it is important to consider these two aspects.

The present study addressed three research questions.

1. To what extent do teacher feedback and ChatGPT feedback impact Chinese university EFL learners' quality of essay revision differently?
2. To what extent do Chinese university EFL learners' PU and PEOU of teacher feedback and ChatGPT feedback differ?
3. How do Chinese university EFL learners perceive the benefits and drawbacks of teacher feedback and ChatGPT feedback?

3. Materials and methods

3.1. Research design, participants, and study context

The present study adopted a within-subject design with a mixed-methods approach, collecting both quantitative and qualitative data. The participants were recruited from a public university in northwestern China. They were 35 first-year university students enrolled in a compulsory English course, which trained students' speaking, listening, reading, and writing skills.

3.2. Instruments

3.2.1. Descriptive writing tasks

Students practiced descriptive writing tasks in the English course. We used two descriptive writing tasks to elicit students' writing, provide feedback, and ask them to revise using the feedback. One task asked students to write on "an unpleasant experience", which received teacher feedback; and the other was on "a person I admire", which received ChatGPT feedback.

The selection of the two tasks was discussed with the participants' English teacher. Both tasks were related to the topics of the two recently completed units in students' textbooks. Hence, students were familiar with the topics and had learned some vocabulary, word collocation, and sentence structures to describe these topics.

3.2.2. Writing feedback

To provide appropriate feedback, the researchers discussed with the English teacher and decided to provide comprehensive and indirect feedback. The comprehensive feedback meant that the feedback covered both meaning and language in writing, consisting of four dimensions: structure, content, language (e.g., grammar, vocabulary, and collocation), and mechanics (e.g., spelling, punctuation, and capitalization). The indirect feedback meant that the English teacher would not provide direct corrections on the writing. Instead, she would only put error indications and metalinguistic cues for the students to locate errors and correct them. The English teacher provided writing feedback on students' essay drafts.

To align ChatGPT feedback and teacher feedback, GPT-4 (OpenAI, 2023) was instructed to generate indirect feedback that only provided error indications and metalinguistic cues and covered the above-mentioned four dimensions. The feedback was required to be presented in a tabular format with a short length (around 30 words for the suggestions of each aspect) to reduce students' cognitive load (The full prompt supplied to GPT is provided in Appendix A). The prompt also included contextual information about the intended recipients, indicating that the student writers were Chinese first-year university EFL learners. Because of the restriction of using ChatGPT in China, the research team obtained feedback from ChatGPT for each student using the same prompt and delivered it to students in Word format. While students lost opportunities to actively interact with ChatGPT, it mimicked the non-interactive feature of teacher feedback.

3.2.3. The questionnaire

We used a 16-item Likert scale questionnaire to examine students' perceptions. The questionnaire represented four scales: namely PU of teacher feedback (4 items, α=.870), PU of ChatGPT feedback (4 items, α=.895), PEOU of teacher feedback (4 items, α=.758), and PEOU of ChatGPT feedback (4 items, α=.786). The PU and PEOU scales were adapted from the technology acceptance literature (Venkatesh & Davis, 2000). The original scales were developed and validated to assess users' PU and PEOU of information systems at work. The scales have been subsequently adapted in various technology-enhanced learning and teaching contexts, including gamified language learning (Luo, 2023), mobile learning (Ali & Warraich, 2024), e-learning (Han & Guo, 2025), and using GenAI in teaching (Dehghani & Mashhadi, 2024). Specifically related to our study, Wang and Han (2022) have adapted and used the four scales to compare PU and PEOU of automated feedback vs. teacher feedback in among Chinese EFL learners, suggesting that these scales were appropriate to be used with Chinese EFL learners. The adapted questionnaire was also reviewed by the English teacher of the participants and were piloted with 12 Chinese EFL learners in the same university where the participants were recruited. Ambiguity in language was revised accordingly. The details of the scales were presented in Appendix B.

3.2.4. Semi-structured interviews

Fifteen semi-structured interviews were also conducted to gain an in-depth insight of students' perceptions of using both types of feedback. Two questions were used to elicit students' responses: 1) "In the revision process, how did you perceive the benefits and drawbacks of teacher feedback?" 2) "In the revision process, how did you perceive the benefits and drawbacks of ChatGPT feedback?"

The semi-structured format of the interviews allowed for open and flexible discussions, giving interviewees the freedom to shape their responses and ample time to reflect and elaborate. Probing questions, such as "What do you mean by…?" and "Could you elaborate…further please?" were used to encourage interviewees to expand on their ideas and clarify their thoughts. All the interviews were audio-recorded and transcribed for analysis.

3.3. Ethical consideration

Prior to data collection, the study received approval from the Human Ethics Committee of the first researcher's university (2024/056). All students in the English class were provided with a participant information statement and a consent form. The information statement clarified the following points: 1) participation in the study was entirely voluntary, 2) their choice to participate or not would not have any impact on their course marks, and 3) they could withdraw from the study at any time. Informed consent was obtained before the data collection.

3.4. Data collection procedure

Data was collected in two English classes in two consecutive weeks. In the first week, students were instructed to write two 200-word descriptive essays on "an unpleasant experience" and "a person I admire". Then English teacher and ChatGPT provided writing feedback on the drafts of "an unpleasant experience" and "a person I admire" respectively. In the second week, students used teacher feedback and ChatGPT feedback to revise the two essays. Upon completion of the revision, students filled out the perceptions questionnaire. To minimize human bias in essay evaluation, students' initial draft essays and the revised essays were graded using a web-based automatic writing evaluation tool, Pigaiwang, which is widely employed by Chinese higher education institutions for assessing English writing (Wang & Han, 2022). The reliability and validity of Pigaiwang have been confirmed through calibration with a large corpus of human-scored English essays (Wang, 2016; Yang & Dai, 2015).

3.5. Data analysis

Data analysis was carried out in SPSS 29. Data analysis began with a screening of the means and standard deviations of all the variables through descriptive statistics (Table 1). Shapiro-Wilk tests and visual assessment of QQ plots were used to examine the normal distribution of the data and residuals. The results of the Shapiro-Wilk (Table 2) and QQ plots (Appendix C) jointly suggest that all the variables met the normal distribution assumptions. To check if students had similar baseline draft essay scores of the two writing tasks, a repeated measures ANOVA was carried out. The results show that the draft essay scores on the two writing tasks did not differ: (F (1,34) = .018, p = .895, partial η^2 = .001), suggesting that the writing tasks did not affect students' writing performance.

To answer the first and second research questions, repeated measures ANOVAs were performed. To answer the third research question, thematic analysis with an interpretive paradigm (Braun et al., 2019; Thorne, 2008) was employed to analyse the interview responses. Thematic analysis was carried out as follows. In the first round of reading, we read the responses repeatedly to determine the qualitative breadth and depth of all the responses. In the second round of reading, we located and highlighted the key features about benefits and drawbacks in each student's response. By focusing on the key features, patterns of responses were more easily identified, which allowed us to identify initial themes. We then compared and contrasted similarities and differences in the themes, generating a defined initial set of categories. Through an iterative process, we modified the definitions of themes so that each stood distinctively from other themes.

4. Results

4.1. Descriptive statistics

Descriptive statistics for all the variables are presented in Table 1.

4.2. Results of the first research question – the extent to which quality of essay revision differed by feedback type

The results of repeated measures ANOVA showed that students obtained significantly higher scores on revised essays using ChatGPT feedback than using teacher feedback (F (1,34) = 18.560, p < .001, partial η^2 = .353). Considering no difference on students' baseline essay draft scores but higher scores of the revised essays using ChatGPT feedback, these results suggested that ChatGPT feedback had a better effect on quality of essay revision in FL writing.

4.3. Results of the second research question – the extent to which PU and PEOU of teacher feedback and ChatGPT feedback differed

Repeated measures ANOVAs showed that students had significantly higher ratings on PU and PEOU of teacher feedback than ChatGPT feedback (PU: (F(1,34) = .4.238, p = .047, partial η^2 = .111); and PEOU: (F (1,34) = .5.338, p = .027, partial η^2 = .136), suggesting that students perceived teacher feedback was more useful and easier to use than ChatGPT feedback.

4.4. Results of the third research question – Chinese university EFL learners' perceptions of the benefits and drawbacks of the two types of feedback

Five major themes were identified on Chinese university EFL learners' perceived benefits and drawbacks of teacher and ChatGPT feedback. These themes revealed that students held mixed feelings towards the two types of feedback, commenting on both positive and negative aspects of the two types of feedback.

4.4.1. Theme 1: Emotional elements in teacher feedback versus emotionless in ChatGPT feedback

Students consistently perceived that the two types of feedback had distinct emotional impacts on their revision process. They felt that the language in teacher feedback carried emotional elements, which encouraged them to revise and sustained their motivation in the process. In contrast, students perceived that ChatGPT feedback lacked human elements and emotions. The following quotes are some examples to demonstrate this theme.

Quote 1: "The teacher's red-pen markings on my essay gave me a sense of human connection. In contrast, the tone of ChatGPT feedback was rather cold, which reduced my motivation to revise."

Quote 2: "The encouraging words in the teacher feedback warmed my heart. I was motivated to improve on these problems because I could feel the effort behind teacher's comments. To me, ChatGPT feedback was cold and emotionless."

4.4.2. Theme 2: Conciseness of teacher feedback versus comprehensiveness of ChatGPT feedback

Students perceived that teacher feedback was concise, but it did not cover many aspects in English writing. In contrast, they praised the comprehensiveness of feedback for addressing even subtle issues, which were often overlooked in teacher feedback. Below are some examples selected from students' interviews.

Quote 1: "While my teacher focused on clarity in my essay, ChatGPT caught subtle issues that teacher overlooked, from word choice to mechanic errors."

Quote 2: "ChatGPT analysed my essay thoroughly, flagging even minor passive voice overuse. My instructor's feedback prioritised

Table 1
Descriptive statistics

variable	teacher feedback				ChatGPT feedback			
	Min	Max	*M*	*SD*	Min	Max	*M*	*SD*
Draft	57.50	87.00	78.90	6.00	61.50	89.50	79.06	7.08
Revision	47.50	87.50	77.07	8.57	65.00	90.50	82.83	5.63
PU	0.00	5.00	3.76	0.93	0.00	5.00	3.31	1.28
PEOU	0.00	5.00	3.55	0.86	0.00	4.50	3.13	1.16

Table 2
Results of Shapiro-Wilk tests

variable	teacher feedback				ChatGPT feedback			
	W	df	*p*	interpretation	W	df	*p*	interpretation
Draft	.98	34	.82	normal	.97	34	.64	normal
Revision	.99	34	.99	normal	.99	34	.96	normal
PU	.97	34	.54	normal	.95	34	.18	normal
PEOU	.98	34	.70	normal	.97	34	.42	normal

critical procedure errors but missed these linguistic details due to time constraints."

4.4.3. *Theme 3: Short and easy language of teacher feedback versus long and complicated language of ChatGPT feedback*

Students perceived that the language used in teacher feedback was easy to comprehend, whereas ChatGPT feedback was often long and difficult to understand. The following quotes from students provided some evidence.

Quote 1: "Reading ChatGPT feedback was just like reading a long English article. Processing the language was quite tedious and effortful for me. I just prefer teacher feedback which is much shorter."

Quote 2: "Although ChatGPT provides feedback on almost every aspect on my writing, it was so lengthy and overwhelming to me. Understanding the language of teacher feedback is much easier."

4.4.4. *Theme 4: Personalization of teacher feedback vs generic format of ChatGPT feedback*

Students appreciated the personalized feedback provided by their English teacher, as they felt their English teacher could understand their thoughts, unique insights, and knowledge gaps. Students disliked the generic format of ChatGPT feedback and felt that it was unnecessary for ChatGPT to provide detailed feedback on every aspect in their writing.

Quote 1: "I felt that in addition to pointing out grammar and spelling errors in my essay, the teacher also paid much attention to my thoughts, the unique insights and creative expression, etc., which allowed her to provide personalized feedback and one-on-one suggestions on how to improve my essay. ChatGPT provides feedback by equally distributing the comments on the four aspects: content, structure, language, and mechanics. I felt that it was unnecessary to focus on every aspect in details as this generic format created difficulties for me to see the key points."

Quote 2: "Because our English teacher understands each student's strengths and knowledge gaps, the comments given by her are always combined with my recent learning situation, which to me, are very appropriate. ChatGPT uses a generic format and similar language expressions to provide feedback to every student, which looks very mechanistic.''

4.4.5. *Theme 5: Teacher's ability to understand the implied meanings in students' writing versus ChatGPT's failure to recognize them*

Students also valued teacher's strengths of understanding the implied meanings in their writings at the same time identified critical gaps in ChatGPT's ability to process the implied meanings.

Quote 1: "When I wrote about my grandmother's silence suggesting her disapproval of the proposal, ChatGPT treated it as ambiguity and suggested to deleting this part as it failed to recognise the cultural practice of conveying dissent through silence. However, my teacher always understands the implied meaning I want to express."

Quote 2: "My metaphor of 'sailing against the current' was marked as 'illogical' by ChatGPT as it failed to grasp the implied and connotated meaning behind it. My teacher never fails to understand.

5. Discussion

The current study compared: 1) teacher feedback and ChatGPT feedback on Chinese university EFL learners' quality of essay revision; 2) PU and PEOU of these two feedback types; and 3) students' qualitative perceptions of the benefits and drawbacks of teacher feedback and ChatGPT feedback.

5.1. *The effect of teacher feedback and ChatGPT feedback on Chinese university EFL learners' quality of essay revision*

Regarding the effect of the two types of feedback on Chinese university EFL learners' quality of essay revision, it was observed that when students used ChatGPT feedback to revise their essays, they attained significantly higher scores than when they used teacher feedback to revise. While previous research has reported the heightened effectiveness of ChatGPT feedback compared to no feedback (Myer et al., 2023) and peer feedback (Mun, 2024) in the process of EFL learners' proofreading and revising, we added extra evidence of more positive impact of ChatGPT feedback over teacher feedback.

The better effect of ChatGPT feedback in improving students' revision quality may potentially be attributed to the comprehensiveness of ChatGPT feedback, which provided various ideas and expansions, such as suggestions of how to better organize the essay, options for different word choice and collocations (Mun, 2024).

This finding aligned with previous research reporting the comprehensiveness of ChatGPT feedback. For instance, Dai et al.'s (2023) compared instructor feedback and ChatGPT feedback on university students' assignments and found that ChatGPT provided more detailed feedback with features of knowledge elaboration and expansion. Concurring with Dai et al. (2023), Guo and Wang (2023) also found that ChatGPT generated a significantly greater volume of feedback than teachers for EFL learners' argumentative writing. Guo and Wang (2023) further pointed out that unlike teacher feedback, which predominantly emphasized content and language issues, ChatGPT distributed more evenly across content, language, and organization. ChatGPT advantage of providing suggestions on how to more appropriate structure the text was also reflected in the semi-structured interviews with our participants. This concurred with our participants' qualitative comments on the usefulness of the ChatGPT feedback for them to rearrange the structure of their essays.

5.2. *Chinese university EFL learners' PU and PEOU of teacher feedback and ChatGPT feedback*

Concerning students' PU and PEOU of the two types of feedback, it was found that students regarded teacher feedback as more beneficial and user-friendly than ChatGPT feedback. A number of comparative studies on students' perceptions of teacher and ChatGPT feedback have shown similar results. For instance, Nazaretsky et al. (2024) reported that when the source of feedback was disclosed, students exhibited a stronger preference for teacher feedback and assigned a lower evaluation to ChatGPT feedback. By comparing students' perceptions and preferences regarding the efficacy of teacher feedback, ChatGPT feedback and peer feedback, Solovey (2024) found that students perceived teacher feedback as the most effective and preferred teachers as the primary source to receive feedback followed by a combination of teacher and ChatGPT in delivering feedback.

Multiple factors might contribute to our findings of students' more positive perceptions of teacher versus ChatGPT feedback. First, our participants favoring teacher feedback might pertain to Chinese students' dominant belief of the authority of teachers. In the Chinese educational context, teachers play an important role in students' learning. Students harbor a high level of trust in teachers, firmly believing in their teaching approaches (Li, 2012). Additionally, Chinese traditional moral values emphasize respect for teachers, which may, to some extent, have strengthened trust in teachers (Sun, 2004), and consequently heightening students' acceptance of words from teachers, including feedback from teachers on writing.

Furthermore, the characteristics of our participants might be another factor of their better perceptions of teacher feedback. Having recently graduated from high school, students lacked prior exposure to interacting with technologies in their learning (including AI) due to the predominant face-to-face teaching in Chinese schools (Li et al., 2021), Students' lack of experience of seeing feedback generated by AI tools might negatively affect their acceptance of ChatGPT, potentially leading to their reservations regarding the usefulness and usability of ChatGPT feedback.

Additionally, students' English proficiency might also affect their abilities to fully utilize the extensiveness of ChatGPT feedback, which also revealed in participants' comments on the challenges of comprehending ChatGPT feedback. Previous research has shown that the feedback generated by ChatGPT was substantially longer, averaging 1335.5 words in contrast to 335.5 words from human (Tzirides et al., 2023). For the first-year university EFL learners, this might induce cognitive overload in students and potentially overwhelm students (Barrot, 2023). Although we limited the length of ChatGPT feedback, we did not equalize the length between ChatGPT and teacher feedback, hence, students' perceptions of the two types of the feedback might be influenced by the differences of the length of the two types of the feedback.

Another factor which might contribute to more negative perceptions of ChatGPT feedback could be the way students interacted with ChatGPT in the feedback process. In this study, our participants did not have opportunities to actively communicate with ChatGPT, such as posing follow-up questions or asking for clarifications; but only received ChatGPT feedback delivered in electronic format to them. This passive feedback process could potentially diminish students' motivation to utilize ChatGPT feedback, which in turn induced their negative perceptions of PU and PEOU of ChatGPT feedback (Escalante et al., 2023).

5.3. Chinese university EFL learners' perceived benefits and drawbacks of teacher feedback and ChatGPT feedback

Inquiring about the Chinese university EFL learners' perceived benefits and drawbacks of the two types of feedback in the revision processes, the semi-structured interview revealed that students exhibited mixed feelings towards both types, recognizing that each feedback having its own strengths and limitations. Students noted that teacher feedback incorporated emotional elements, a crucial aspect in motivating them to revise their essays. This aligned with previous research showing the pivotal role of teacher-student interaction in students' responses to feedback (Lee & Schallert, 2008). These findings may suggest that when using ChatGPT to provide feedback to EFL learners' writing, it is important to supplement the emotional limits of AI tools by the human elements in the process.

Furthermore, in line with Teng (2024)'s findings, our participants also recognized the strength of the comprehensive nature of ChatGPT feedback, which might contribute to its heightened effectiveness on students' higher quality of revised essays. While ChatGPT exceled in detail, students consistently noted its exhaustive length and complicated language, which made students feel overwhelmed and added extra cognitive load for students to process the feedback in the revision process. This perceived drawback of ChatGPT feedback might contribute to students' lower ratings on the PEOU of ChatGPT feedback than teacher feedback. One possible reason for students' negative perceptions of the comprehensible nature of ChatGPT could be linked to their low-level of English proficiency. This finding highlighted that language proficiency should be an important factor to consider when integrating ChatGPT into the feedback process.

One unique insight revealed from students' qualitative comments concerned with ChatGPT's inability to comprehend some implied meanings in students' expressions, such as the Chinese way of expressing disapproval from silence. This finding suggested that for potential value of developing cultural adaptive GenAI systems by integrating culturally specific values into the AI training process.

5.4. Limitations and directions of future research

Despite some interesting findings, this study has several limitations, which affects the generalizability of the research results. First and foremost, the feedback source (teacher vs. ChatGPT) was fixedly paired with a specific writing topic. Although we established the baseline equivalence of students' draft essay scores on the two writing topics, fixed topic with feedback type still remains a potential source which might confound the effect of feedback type on the quality of essay revision. Future research should address this limitation by employing fully counterbalanced or randomized within-subject designs. Second, while we required ChatGPT to give relatively short feedback (see Appendix A for the prompt for ChatGPT), such requirement was not imposed on the English teacher when providing writing feedback. As a result, ChatGPT tended to produce feedback with relatively equal length with each suggestion being around 30 words. However, the feedback from the English teacher varied in terms of the length, with some being detailed and lengthy whereas others being brief and short. These discrepancies between ChatGPT and teacher feedback might cast different cognitive load in students' revision processes, and might also affect students' perceptions of ease of use and their recognition of the strengths and limitations of the two types of feedback. Future studies should ensure that the length of teacher feedback is standardized and to be matched with ChatGPT feedback to eliminate the potential influence of feedback length on quality of revision and students' perceptions. Third, participants were recruited from a single proficiency level (first-year university students) and a single institution, affecting generalizability. Expanding the sample to include students from diverse institutions, proficiency levels, and measuring students' prior AI experience are crucial next steps in future research.

5.5. Pedagogical implications

Although our findings suggested that ChatGPT feedback outperformed teacher feedback in enhancing EFL learners' quality of revision, students nevertheless cherished user-friendly features and positive emotional elements carried by the feedback provided by teachers. Therefore, it is recommended that EFL teachers consider using ChatGPT feedback as a supplement to enhance the feedback process rather than relying on it entirely. EFL instructors can integrate ChatGPT feedback into teacher feedback (Asadi et al., 2025). For instance, EFL teachers can harness AI for preliminary evaluations of students' essays, thereby freeing up time to offer more in-depth and personalized feedback. This blended use of teacher and ChatGPT feedback allows students to receive quick identification of surface-level problems in writing (e.g., grammatical errors and mechanics problems) by AI, at the same time students can also embrace emotional encouragement from teachers. Moreover, teachers should familiarize students with ChatGPT, clarifying its features and appropriate usage methods to enhance students' understanding and effective application of this emerging technology in the essay revision process (Strzelecki, 2023; Zou & Huang, 2024). Given that actively interacting with ChatGPT tends to increase students' motivation in using it, in the feedback and revision processes, teachers should encourage students to ask follow-up questions and to seek clarifications, which may enhance students' positive perceptions and adoption of ChatGPT feedback (Kizilcec, 2024).

CRediT authorship contribution statement

Feifei Han: Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Zehua Wang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethical statements

The study was approved by the Human Ethics Committee of the first researcher's university (2024/056). Informed consent was obtained from all participants, and their privacy rights were strictly observed.

Open data statements:

The data can be obtained by sending request e-mails to the corresponding author.

Disclosure statement

The authors report there are no competing interests to declare.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Funding details

This research was jointly supported by the 2023 Project of Shaanxi Provincial 14th Five-Year Plan for Education Grant (Grant No. SGH23Y2571) and the 2025 Special Research Project of Philosophy and Social Sciences in Shaanxi Province (Grant No. 2025HZ0848).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Prompt for ChatGPT

In the following you will read an English essay written by a first-year Chinese university EFL student. Please give feedback on the four aspects: structure, content, language (e.g., grammar, vocabulary, and collocation), and mechanics (e.g., spelling, punctuation, and capitalization). Give feedback in a highly structured manner displayed as a table. The first column lists the four aspects, one row for one aspect. The second column provides suggestions for improvement on the relevant aspect: do not point out the errors in the essay directly, instead, only provide error indications and metalinguistic cues. Each suggestion should be approximately 30 words. The third column provides three examples for improvement on the relevant aspect. The examples must not contain fully formulated sentences but only short aids to revise the essay.

Appendix B. The Scales to Measure PU and PEOU of Teacher and ChatGPT Feedback

scale	item	item description
PU of teacher feedback	1	Using teacher feedback improves my performance in revising my English writing.
	2	Using teacher feedback increases my productivity when revising my English writing.
	3	Using teacher feedback enhances the effectiveness of my English writing revision.
	4	I find teacher feedback to be useful in my English writing revision.
PEOU of teacher feedback	1	My interaction with teacher feedback while revising my English writing is clear and understandable.
	2	Interacting with teacher feedback while revising my English writing does not require much mental effort.
	3	I find teacher feedback to be easy to use for revising my English writing.
	4	I find it easy to use teacher feedback to revise my English writing.
PU of ChatGPT feedback	1	Using ChatGPT feedback improves my performance in revising my English writing.
	2	Using ChatGPT feedback increases my productivity when revising my English writing.
	3	Using ChatGPT feedback enhances the effectiveness of my English writing revision.
	4	I find ChatGPT feedback to be useful in my English writing revision.
PEOU of ChatGPT feedback	1	My interaction with ChatGPT feedback while revising my English writing is clear and understandable.
	2	Interacting with ChatGPT feedback while revising my English writing does not require much mental effort.
	3	I find ChatGPT feedback to be easy to use for revising my English writing.
	4	I find it easy to use ChatGPT feedback to revise my English writing.

Note: These items were randomized when administering the survey.

Appendix C. QQ Plots of All the Variables

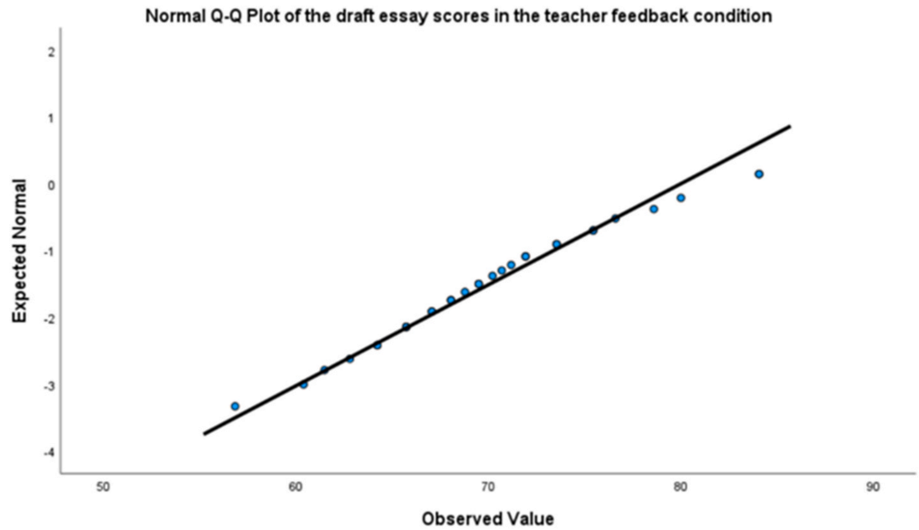

Fig. 1. QQ plots of the draft essay scores in the teacher feedback condition.

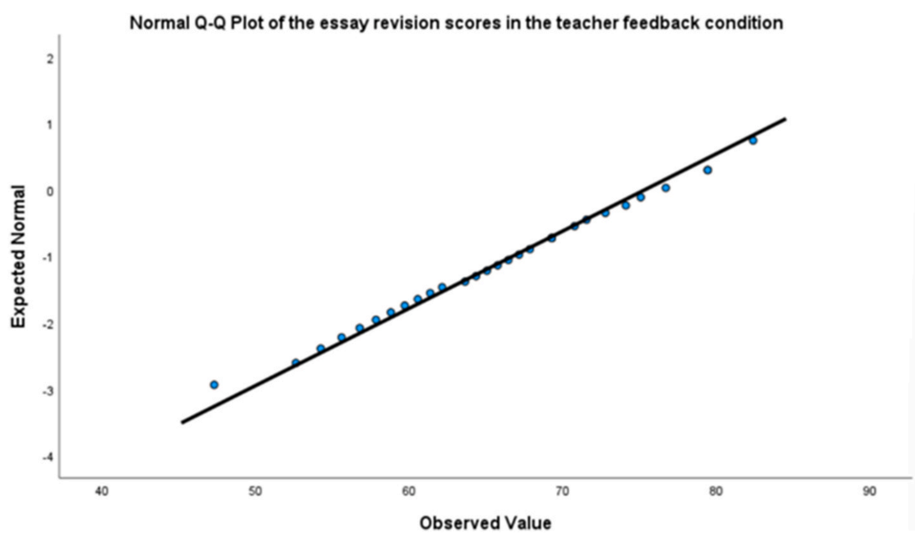

Fig. 2. QQ plots of the essay revision scores in the teacher feedback condition.

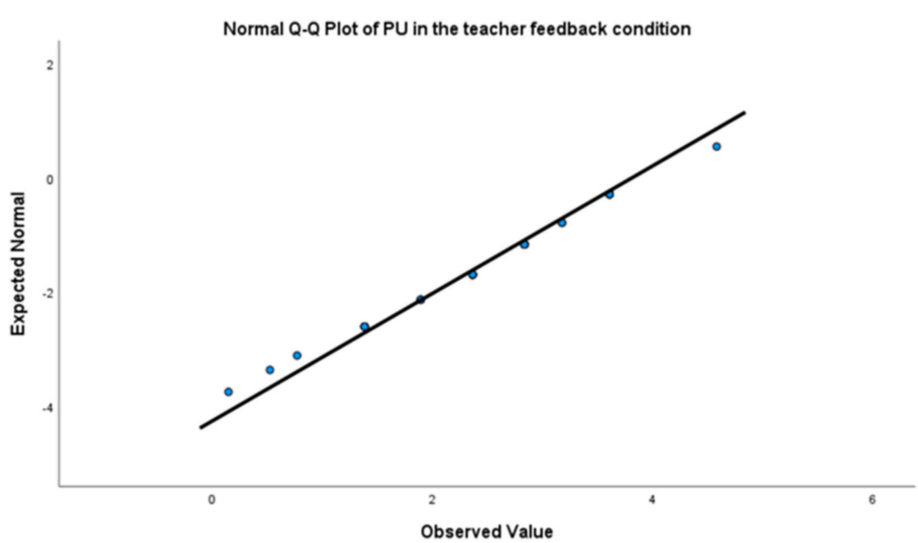

Fig. 3. QQ plots of PU in the teacher feedback condition.

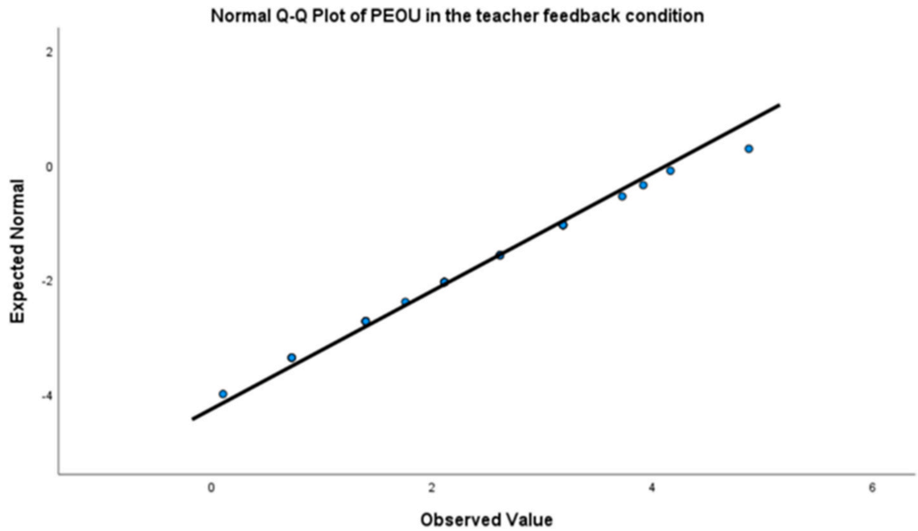

Fig. 4. QQ plots of PEOU in the teacher feedback condition.

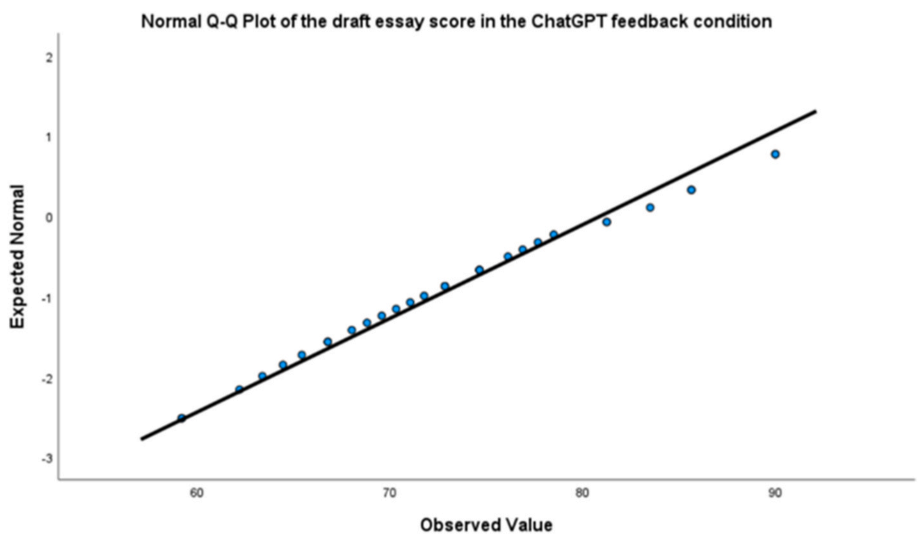

Fig. 5. QQ Plots of the draft essay score in the ChatGPT feedback condition.

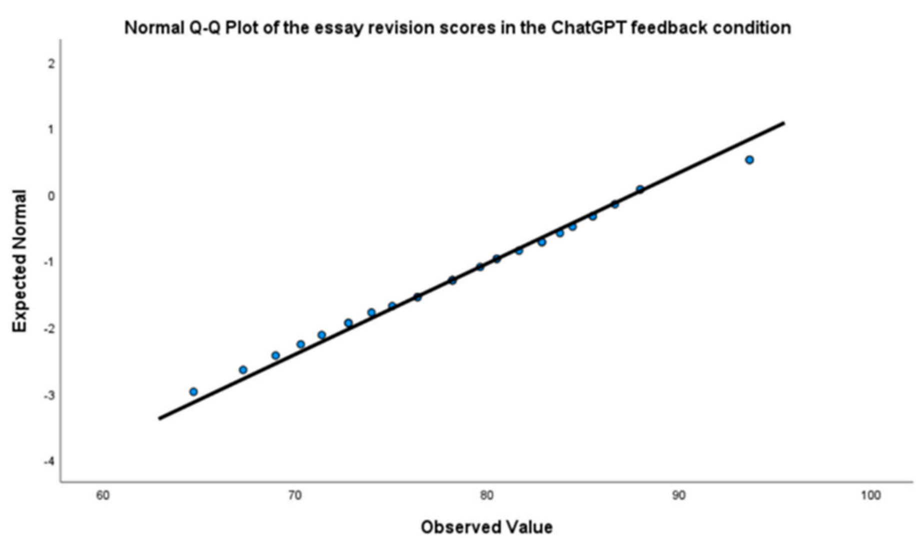

Fig. 6. QQ plots of the essay revision scores in the ChatGPT feedback condition.

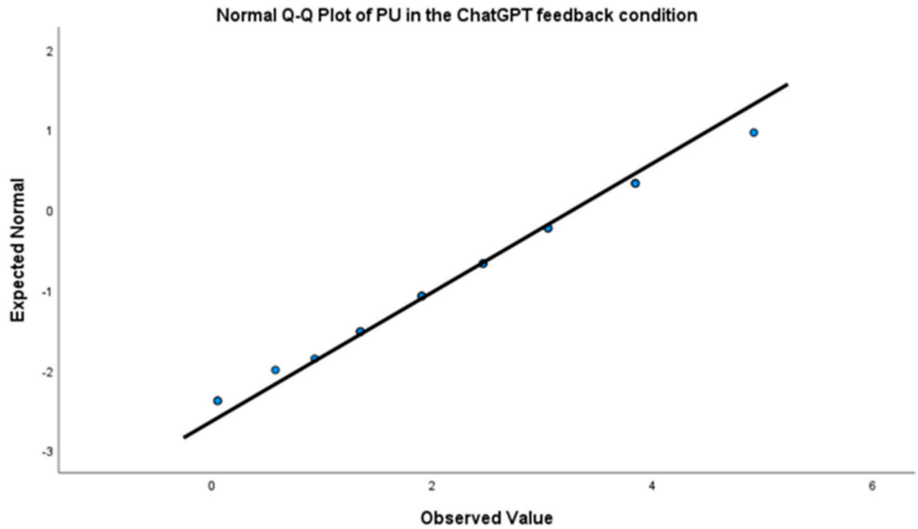

Fig. 7. QQ plots of PU in the ChatGPT feedback condition.

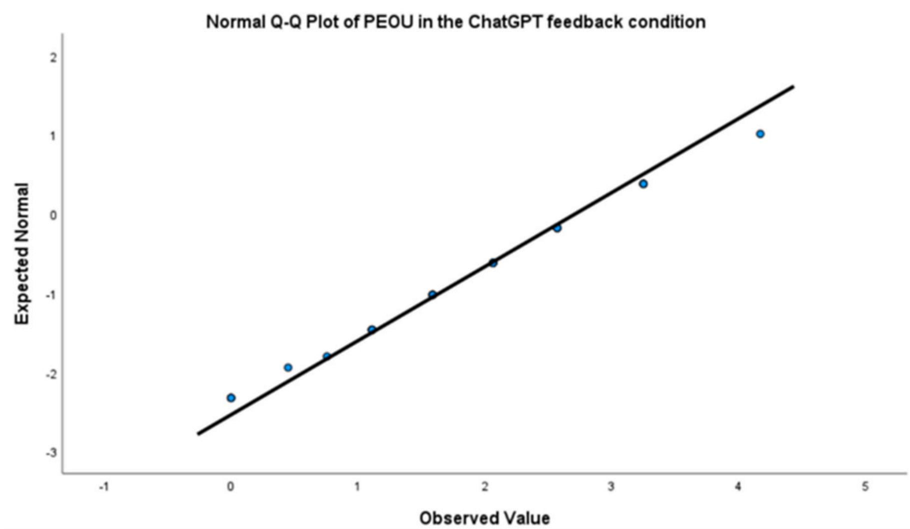

Fig. 8. QQ plots of PEOU in the ChatGPT feedback condition.

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